# An LLM-based Agent Simulation Approach to Study Moral Evolution

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## Abstract

The evolution of morality presents a puzzle: natural selection should favor self-interest, yet humans developed moral systems promoting altruism. We address this question by introducing a novel Large Language Model (LLM)-based agent simulation framework modeling prehistoric hunter-gatherer societies. This platform is designed to probe diverse questions in social evolution, from survival advantages to inter-group dynamics. To investigate moral evolution, we designed agents with varying moral dispositions based on the Expanding Circle Theory (Singer 1981). We evaluated their evolutionary success across a series of simulations and analyzed their decision-making in specially designed moral dilemmas. These experiments reveal how an agent's moral framework, in combination with its cognitive constraints, directly shapes its behavior and determines its evolutionary outcome. Crucially, the emergent patterns echo seminal theories from related domains of social science, providing external validation for the simulations. This work establishes LLM-based simulation as a powerful new paradigm to complement traditional research in evolutionary biology and anthropology, opening new avenues for investigating the complexities of moral and social evolution.

## Introduction

The emergence and evolution of morality represents one of the most enduring puzzles in evolutionary biology and social sciences (Haidt 2007; Greene 2013). From an evolutionary standpoint, natural selection should favor individuals who maximize their reproductive success, often through selfish behaviors that increase resource acquisition at others' expense (Dawkins 1976). Yet, humans and some other species have evolved complex moral systems that frequently promote cooperation, altruism, and other prosocial behaviors that can seemingly contradict individual fitness maximization (Tomasello 2016). This apparent contradiction presents a profound scientific question: Under what conditions does morality provide an evolutionary advantage?

Prior research approached this question via multiple paradigms. Evolutionary game theory showed strategies like reciprocity can outperform selfishness under certain conditions (Nowak 2006; Axelrod and Hamilton 1981). Anthropologists documented moral universals and cultural variations (Henrich 2015; Curry, Mullins, and Whitehouse 2019a). Evolutionary biology proposed mechanisms—kin selection (Hamilton 1964), reciprocal altruism (Trivers 1971), and group selection (Wilson and Wilson 2007)—to explain altruistic trait evolution. Some moral frameworks identified patterns such as the expanding circle of moral concern (Singer 1981), and fundamental moral dimensions (Haidt 2007). While these approaches have yielded valuable insights, they face significant methodological limitations when investigating the whole cognitive and behavioral dynamics of morality evolution. Traditional mathematical models necessarily abstract away the rich complexity of human cognition and interaction with the environment. Evolutionary biologists and anthropological studies provide observations and hypotheses without direct and causal tests. Descriptive moral frameworks illuminate current moral patterns but cannot directly observe their evolution.

Recent advances in LLMs present a novel methodological opportunity to address these limitations. LLM-based agent simulations can model entities with sophisticated cognitive architectures—including values, memory, perception, reasoning, and social dynamics—that generate emergent, complex behaviors (Park et al. 2023; Horton 2023). This simulation paradigm allows us to observe interactions between moral cognition, behavior, and evolutionary outcomes under controlled conditions while providing rich, realistic psychological details that surpass traditional agent-based models (Aher, Arriaga, and Kalai 2023). Nevertheless, the existing methods barely support research on morality evolution.

To address the gaps, our research advances the study of human evolution through three primary contributions: 1) Methodological: We pioneer a computational approach using LLM-based agents to bring psychological realism to evolutionary simulations, overcoming the limitations of traditional abstract models; 2) Programmatic: We release MoRE, an extensible agent framework, and Social-Evol, a versatile environment platform. Together, they enable multifaceted inquiry into social evolution—from norm formation to inter-group conflict—through both long-term and scenario-based simulation; 3) Empirical: We conduct novel experiments to determine how the evolutionary success of various moral dispositions is shaped by environmental pressures, cognitive limitations, and social configurations. Our

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Table 1: Comparison of Existing LLM-based Simulation Frameworks for Social Science.

| Environment | Morality | Memory | Team Formation | Competition | Evolution | Engine |
|---|---|---|---|---|---|---|
| Park et al. (2023) | ✗ | Short/Long-term | ✗ | ✗ | ✗ | LLM |
| Li et al. (2023) | ✗ | Undefined | Multi-round Negotiation | ✗ | ✗ | LLM |
| Horton (2023) | ✗ | Undefined | ✗ | Explicit | ✗ | LLM |
| Aher, Arriaga, and Kalai (2023) | ✗ | Short-term | ✗ | ✗ | ✗ | LLM |
| Wang, Chiu, and Chiu (2023) | ✗ | Short-term | ✗ | ✗ | ✗ | LLM |
| Huang et al. (2024) | ✗ | Short-term | Multi-round Negotiation | Implicit | ✓ | LLM/RL |
| Wang et al. (2024) | ✗ | Short-term | ✗ | ✗ | ✗ | LLM |
| Piao et al. (2025) | ✗ | Short/Long-term | Task-dep. | Task-dep. | ✓ | LLM |
| Guan et al. (2024) | ✗ | Short/Long-term | Multi-round Negotiation | Explicit | ✓ | LLM |
| Dai et al. (2024) | ✗ | Short/Long-term | Negotiation-based | Explicit | ✓ | LLM |
| Ours | ✓ | Short/Long-term | Multi-round Negotiation | Explicit | ✓ | LLM |

simulation's emergent phenomena align with established social science theories, thereby validating our framework's ability to produce meaningful and interpretable results.

## Related Work

**Evolutionary Origins of Morality** Evolutionary biologists have proposed various mechanisms to explain how altruistic traits might evolve. Theories of kin selection (Hamilton 1964) and reciprocal altruism (Trivers 1971) show how limited forms of cooperation could evolve among relatives and repeated interaction partners. Recently, cultural group selection theories (Boyd, Richerson, and Henrich 2011; Henrich 2015) explain how groups with stronger moral norms outcompeted others, leading to the genetic evolution of psychological predispositions supporting moral behavior. Evolutionary game theory provided mathematical frameworks demonstrating how cooperation can evolve under some strategies similar like previously mentioned mechanisms by showing strategies can yield higher payoffs than pure selfishness under specific conditions. Nowak (2006)'s "five rules for the evolution of cooperation" identifies key mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection.

Though these works provide great insights and a mathematical foundation for cooperation and moral evolution, they highly abstract away the rich complexity of human cognition and cooperation using mathematical models, blocking a full view of moral evolution dynamics.

**Moral Frameworks** Moral Foundations Theory identifies five moral dimensions: care/harm, loyalty/betrayal, authority/subversion, and sanctity/degradation (Haidt and Joseph 2007). The Theory of Dyadic Morality (Gray, Young, and Waytz 2012) emphasizes harm as the root of morality, while Morality-as-Cooperation theory (Curry, Mullins, and Whitehouse 2019b) identifies seven cooperative behaviors as essential: helping kin, helping group members, reciprocating, being brave, deferring to superiors, dividing resources fairly, and respecting others' property. Other theories ground morality in distinctions between rules and conventions (Turiel 1983), social-relational models (Fiske 1992; Rai and Fiske 2011), specific moral emotions (Rozin et al. 1999), or an ethic of care (Gilligan 1982).

While existing theories offer rich descriptions of morality's central traits, our initial investigation desires a framework that organizes these traits into scalable "levels" for a more systematic analysis. This led us to The Expanding Circle Theory (Singer 1981) provides an ideal structure for this purpose. Its concept of a "circle of concern" that expands from the self, to kin, and finally to society offers a clear, tiered structure for defining an agent's moral level. Furthermore, this progression provides a natural way to integrate key concepts from other theories: care and loyalty are paramount within the kin circle, while fairness and reciprocity are essential for the stability of the broader group circle. The cross-cultural relevance of this concentric model is highlighted by its deep resonance with philosophical traditions like Confucianism (Fei 1992), which also defines morality through the proper ordering of relational duties. Given its implementable structure and integrative power, we adopt the Expanding Circle as the primary theoretical foundation for our simulation.

**LLM-Based Agent Simulation** Recent advances in large language models (LLMs) provide the methodological foundation for our approach to studying morality's evolution. LLM-based agent simulations can model entities with sophisticated cognitive architectures—including values, memory, perception, reasoning, and social dynamics—that generate emergent, complex behaviors (Park et al. 2023; Horton 2023). LLM agents allow for controlled experimentation with variables that would be impossible to manipulate in real-world settings; they provide transparent access to agents' decision-making processes; and they can simulate long timescales of social development (Piao et al. 2025).

Prior work has demonstrated the versatility of this approach. Park et al. (2023) created an interactive simulated small town where generative agents exhibited complex social behaviors, including relationship formation and collective problem-solving. Horton (2023) applied LLM agents to economic simulations, finding that agents replicate known economic phenomena while providing unprecedented access to reasoning processes. Aher, Arriaga, and Kalai (2023) val-

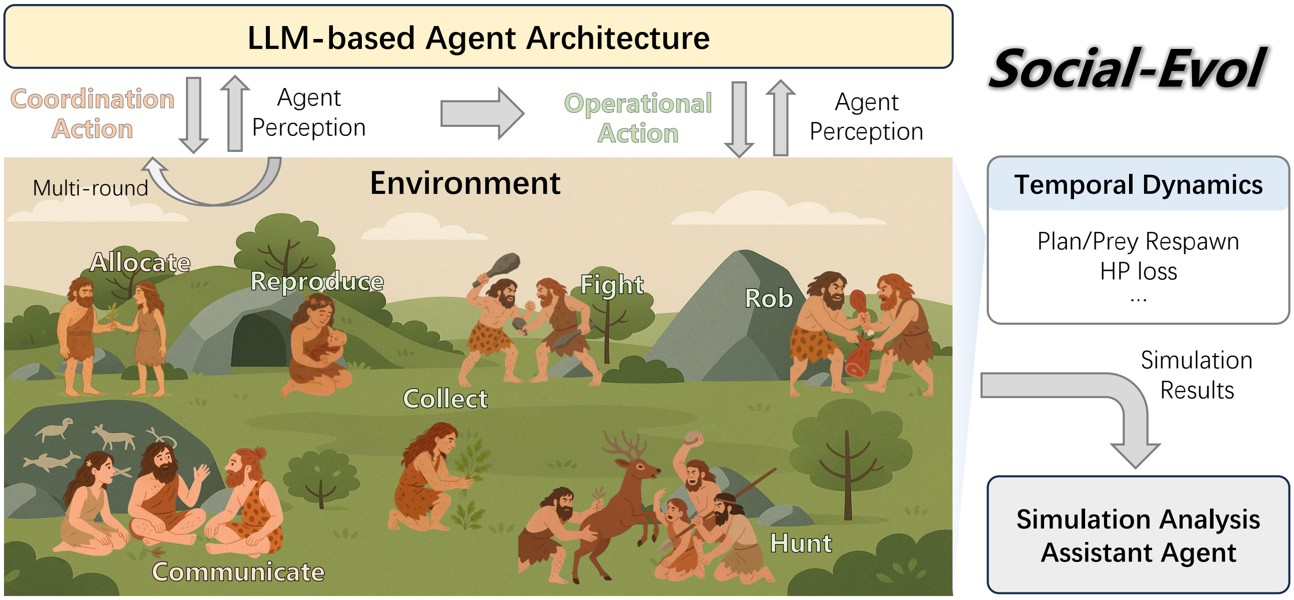



Figure 1: Overview of our simulation framework. The *Environment* follows defined temporal dynamics and iterates between (multi-)coordination rounds and action rounds to interact with agents. The *Agent Cognitive Architecture* includes a moral value module prescribing moral types based on expanding circles of concern, a perception module processing environmental information, and cognitive modules that update memory, form judgments, and generate action plans consistent with the agent's moral type. Before execution, agents perform self-reflection to verify consistency with observed facts and moral dispositions. The *Simulation Analysis Assistant Agent* is a developed tool to automatically analyze simulation results.

idated that LLM agent simulations can reproduce results from human behavioral experiments, showing its potential as a complementary methodology in social science research. However, existing environments lack the complete settings of agent morality, cooperative and competitive interactions, and evolution features, as demonstrated in Table 1. One relevant recent work is "Artificial Leviathan", which explores social order in LLM agent societies, while they assume all agents are inherently selfish and focus on how social order emerges from this assumption (Dai et al. 2024).

Therefore, by explicitly modeling agents with varying moral dispositions to better reflect the diversity of human moral psychology, our work represents the first systematic application of LLM-based simulations to investigate the evolution of morality in prehistoric human societies, where moral systems likely first emerged.

## Agent Cognitive Architecture (MoRE)

Our agent design MoRE is a **m**orality-driven **e**ntity-oriented cognitive processing architecture with **r**eflection capability.

**Agent Traits: Moral Types as Example** To model evolutionary pressures, agents share a foundational value: maximizing survival and reproduction. It's noteworthy that beyond this, the framework supports researchers to customize agent traits to conduct various research problems in social science beyond this paper's scope. In this paper, we focus on the moral simulation problem and implement moral dispositions based on the "expanding circle" concept (Singer 1981), defining a spectrum of morality:

*Self-focused agents* care exclusively about themselves. But in implementation, we notice a definitional challenge: purely selfish agents have no interest in reproduction, but defining them to care about offspring would conflate them with kin-focused agents. Therefore, we define them as investing in reproduction but providing no further offspring care, similar to r-selected species (fish, amphibians, invertebrates) that maximize reproductive success through quantity rather than parental care (Pianka 1970; Stearns 1992; Gross 2005; Reznick, Bryant, and Bashey 2002; Trumbo 2012)

*Kin-focused agents* extend moral concern to genetic relatives, providing care and resources to family members while treating non-kin instrumentally.

*Group-focused agents* extend moral concern beyond kin to include non-related group members. However, defining a group remains a challenge: who constitutes the "group" worthy of moral consideration? We define two variants:

- *Reciprocal group moral agents* extend care only to those who reciprocate similar moral concern, creating a self-consistent moral circle based on mutual recognition.
- *Universal group moral agents* extend care to all individuals regardless of their morality. It is expansive, and its non-violent orientation aligns with some intuitive conceptions of morality. However, this variant presents theoretical inconsistencies—violating fairness and reciprocity principles while benefiting agents who may undermine group welfare, such as selfish agents.

This framework yields four distinct moral types that enable systematic investigation of how different moral dispositions affect evolutionary outcomes. We acknowledge that this discrete categorization simplifies the continuous nature of moral concern in humans for experimental tractability. More importantly, this demonstrates the ability of the framework to customize agent traits by defining the characteristics under the aiming social science research problems.

**Agent Cognition Framework** Our simulation employs agent-based modeling powered by LLM to capture sophisticated cognitive and social dynamics. Typical agent cognition processes are as follows:

*Agent Initialization* Agents are initialized with (1) a profile containing value characteristics aligned with designated moral type; (2) environmental rules governing the simulation; and (3) a knowledge handbook containing a commonsense understanding of environmental dynamics and causal relationships. It ensures agents begin with a comparable baseline understanding without artificially constraining their decision space. Importantly, no agent receives privileged strategic information, preventing methodological bias.

*Perception Module:* This component processes current status information about plants, prey, and other agents (e.g., HP, age) along with recent activities up to a configurable number of past steps, mirroring human short-term memory.

*Cognitive Processing System:* We designed an integrated entity-based system that maintains memory, makes judgments, and forms dispositions around entities like other people and hunting animals. This is in contrast to the event-based cognitive processing that records a log-book-like memory and decision history. Our preliminary studies demonstrate that this method effectively prompts LLMs to consider relevant context and perform appropriate reasoning compared to simpler approaches. The entity-based structure provides a template for identifying important information and creating a narrative-like understanding.

*Action Planning:* This module prioritizes updated memories and dispositional plans to formulate specific actions. This is crucial because the simulation environment may contain many entities toward which an agent might have multiple intended interactions.

*Reflection Module:* This verification component ensures cognitive processing and action planning remain consistent with factual information and faithful to the agent's moral type, while producing properly formatted responses.

## SOCIAL-EVOL Framework

SOCIAL-EVOL (Fig. 1) is a text-based prehistoric hunter-gatherer society with resource constraints, social dynamics, and environmental challenges, enabling research on evolutionary pressures that likely shaped early human morality.

### Environment

**Survival Mechanics** Agents maintain health points (HP) that must remain above zero to sustain life. HP diminishes gradually over time, representing basic metabolic costs, and decreases substantially from injuries sustained during hunting or conflict. Agents replenish HP by collecting plants or hunting prey, with successful actions immediately increasing their HP. Additionally, agents face a maximum lifespan constraint that eventually results in death regardless of HP management, modeling natural senescence.

**Resource Setting** The environment contains two primary resource types: plants and animals. Plants represent low-risk, low-reward resources that agents can reliably collect. Animals offer high-risk, high-reward resources that yield more HP but present significant acquisition challenges. Plants are configured with initial quantity, capacity, nutrition value, and respawn delay. Animals are configured with HP, physical ability, and respawn interval. Those settings can be used to configure the abundance of the available resources.

**Production and Reproduction Mechanics**

*Hunt, Collect:* Agents collect plants solely to gain HP without risks whenever plants are available. Hunting allows collaboration, which embodies realistic risk-reward trade-offs: success probabilities and damage inflicted depend on agent-prey physical ability differentials. Failed hunts cause automatic counter-attacks (i.e., HP loss) to agents; successful attempts progressively reduce prey health until death. This difficulty incentivizes collaboration, improving success rates and distributing risks.

*Reproduce:* Agents meeting age and HP thresholds can reproduce, though reproduction imposes significant metabolic costs. Offspring begin life with minimal HP, creating vulnerability that typically requires parental investment (food sharing) to ensure survival.

**Social Interaction Mechanics** The environment supports multiple forms of social interaction:

*Allocate:* Agents can voluntarily transfer HP to others, enabling cooperative behaviors and care-based relationships.

*Communicate:* Agents can exchange messages with specific individuals or groups simultaneously. This communication system enables coordination, information sharing, and relationship development.

*Rob, Fight:* Agents may attempt robbery to steal HP or fight to inflict damage without HP gain. Success probability and outcomes depend on relative physical abilities, with gains proportional to the aggressor's strength. Such actions have slightly elevated HP costs due to their intensity. Unlike hunting, failed aggressive attempts do not trigger automatic retaliation; victims must initiate responses independently. This design enables agents to freely express their moral strategies without artificial constraints.

This environmental design creates a complex adaptive system where survival pressures, resource competition, cooperation opportunities, and communication capabilities interact to influence the differential success of varied moral dispositions. Detailed rule explanations and configuration settings appear in the supplement.

### Simulation Operating Cycle

The simulation operates as a sequential process where agents and the environment interact in defined steps: 1) *Environment Update*, where the simulation refreshes resource availability, agent status changes, and advances time; 2)

*Agent Perception*, where each agent receives observations about the current environmental state and recent activities; 3) *Cognitive Processing*, where agents use their architecture to process perceptions, update memory, form judgments, and develop dispositional plans consistent with their moral type toward different entities (prey or other agents) or goals (reproduction etc.); 4) *Action Planning*, where agents need to consider their dispositional plans and conditions to prioritize and make specific action plans for the next few steps. Note that each simulation step includes one or more coordination rounds (allocate/communicate) and one action round (reproduce/collect/fight/rob/hunt), and the environment update itself with defined temporal dynamics; and 5) *Consequence Resolution*, where outcomes of all actions are determined. This cycle repeats continually, enabling emergent complex social behaviors while maintaining tractable simulation parameters. The LLM serves as the cognitive engine for each agent, providing reasoning capabilities necessary to navigate moral dilemmas, form social strategies, and respond to environmental pressures in ways that reflect human-like cognitive processes.

## Two Supported Game Modes

Our environment allows two simulation game modes to study both long term evolution and specific decision dynamics under targeted scenarios.

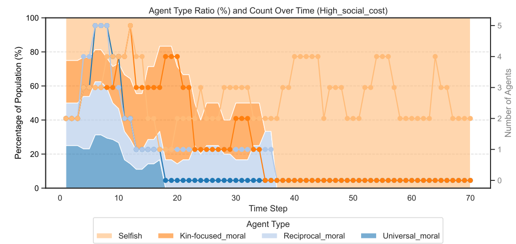
**Evolutionary Game** The evolutionary games are full simulations of agents' behavior until all agents die or reach maximum steps. This game captures the long-term, emergent outcomes of moral evolution.

**Mini Games** This game setting is designed to isolate a crucial step in the causal chain from morality to fitness: the moment an action is chosen. By placing agents in a specific, controlled scenario like moral dilemmas, we can clearly observe how different moral dispositions translate into distinct behaviors, thus illuminating a key mechanism that drives the broader dynamics seen in the full simulation.

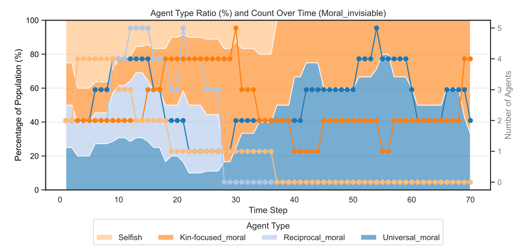
## Simulation Data Analysis Assistant

Throughout our project development, we identified a significant challenge in LLM-based agent simulations: interpreting the vast quantities of generated data. While having rich, multidimensional data offers tremendous analytical potential, extracting meaningful insights from this complexity requires specialized methodological approaches. To address this challenge, we developed a simulation analysis assistant agent that serves two critical functions. First, it automatically generates comprehensive statistical reports containing the key metrics visualized in our figures. Second, we implemented a series of function calls to enable an interactive Q&A ability when user uses a readily available code copilot agent like Copilot or Cursor. It can allow researchers to interrogate specific agent behaviors, motivations, and decision processes (e.g., "Why did Agent X perform action Y?"). This analytical tool has proven invaluable for understanding simulation dynamics and iteratively refining our agent design architecture. We provide detailed specifications of this system in the supplement.


(a) Scenario with major baseline setting




(b) Scenario with high social interaction cost




(c) Scenario with invisible moral type

Figure 2: Agent populations and ratios of four moral types

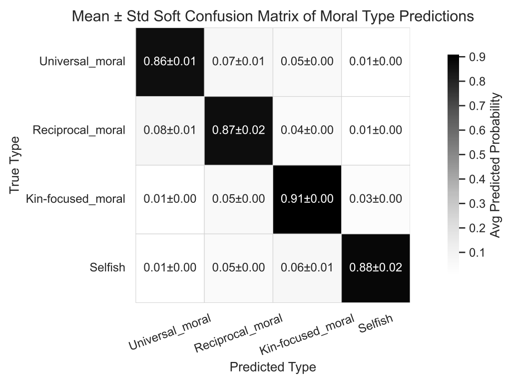
# Experiment

## Evolutionary Game

All the agent simulation runs are using OpenAI's GPT-4.1-mini API. Here we present our baseline simulation:

**Moral Evolution** We run several exemplary experiments to demonstrate how moral dispositions interplay with various settings. Our baseline configuration employs a non-scarce resource, low social interaction cost, and direct moral type observability to provide a relatively easy mode of survival. In another two experiments, we change the setting to **High Social Interaction Cost** and **Invisible Moral Type**, respectively. Each simulation is initialized with 8 agents, where 2 agents per moral type. The progression of the population ratio of each moral type is shown in Fig. 2. In the baseline setting, kin-focused moral agents ultimately dominate the population. However, when interaction costs are high—making collaboration more difficult—selfish agents take over. In a third condition, where agents cannot directly observe each other's moral types and must infer them through interaction, both kin-focused and universal moral agents tend to survive until the end. A closer analysis of the underlying dynamics reveals that kin-focused agents outperform others in the baseline setting because they avoid conflicts over resource distribution and rapidly produce off-


(a) HP trajectories of example agents.




(b) Confusion matrix of moral type inference.

Figure 3: Validation results of the baseline simulation.

spring early in the simulation. In the high-interaction-cost condition, more moral agents spend excessive time negotiating collaboration terms rather than engaging in productive activities like hunting or gathering, allowing selfish agents to exploit this delay and gain an advantage. In the setting with hidden moral types, kin-focused agents continue to benefit from tight in-group coordination. Meanwhile, universal moral agents also persist—not because of superior collaboration efficiency, but because their strict rejection of all violence, including justified revenge, minimizes the risk of misinterpretation. In contrast, reciprocal moral agents, who selectively punish defectors, are often mistaken for selfish individuals and therefore suffer from social exclusion or retaliation. Additional experiment (**Resource Scarcity**) and detailed results can be found in the Appendix.

**Simulation Validation** To showcase the effectiveness of the simulation, we validated the results from the perspectives of environmental feedback and consistency between morality and behavior. For the environment feedback, we randomly selected one agent from each moral type in the baseline experiment, plotting their HP curves alongside their actions in Fig. 3 (a). It reflected the correct HP changes caused by various actions. For the morality analysis, we took the kin-focused agent, who illustrates strong familial bonds and self-sacrifice, as an example. Born at step 15 with 3 HP, it gained HP through two big lucky collections, then delivered two children, and frequently shared resources. After another reproduction, however, it did not gain enough energy in time and died. The regular HP fluctuations show a balance between self-preservation and family support. Its life was characterized by frequent communication with family members, seeking alliances for mutual survival, and prioritizing children's well-being over personal gain.

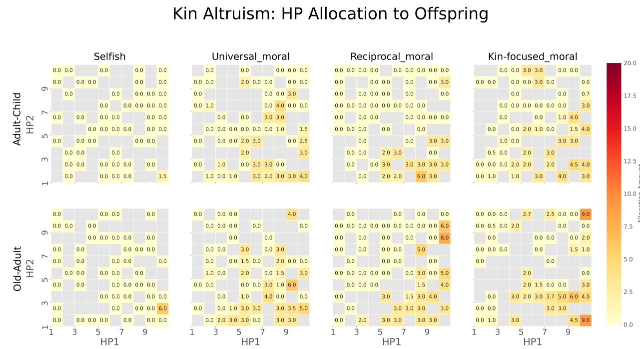



Figure 4: Kin Altruism: A Minigame on Family Resource Allocation.

To further verify the morality-behavior consistency, we applied advanced models (GPT-4.1) to perform moral type inference from observed behaviors. Averaging three inference trials yielded a confusion matrix shown in Fig. 3 (b). The inference achieved high accuracy, with the standard deviation under 0.02, showing the reliability of our simulation.

## Mini-game

We constructed an example mini-game to study how morality interacts with other factors for an resource allocation scenario within family. More mini-game examples are in the Appendix.

To understand how moral dispositions shape resource sharing within families, we studied intergenerational transfers between parents and children. Our experiment involved parent-child dyads from two distinct life stages: young parents with infants and elderly parents with adult children. We modeled four moral dispositions—reproductive selfishness, universal group-focus, reciprocal group-focus, and kin-focus—creating eight unique scenarios. For each, we generated a heatmap to visualize parental transfer behavior. In these maps, the x- and y-axes represent the parents' and child's health (HP), respectively, while the color indicates the amount of resources transferred. The results reveal that provisioning strategies are systematically driven by moral type, showing clear thresholds for initiating aid, different sensitivities to self versus offspring health, and complex interactions between moral orientation and age.

As shown in Fig. 4, parents generally transfer surplus health points (HP) to their children. This behavior is strongly modulated by age, with older parents allocating significantly more resources as they approach their lifespan limits. Moral dispositions, however, introduce the most striking variations. For instance, reproductively selfish parents tend to hoard resources for self-preservation, while in stark contrast, kin-focused parents consistently prioritize their offspring. In extreme cases, these kin-focused parents sacrifice nearly all of their own HP to ensure their child's survival and fitness.

## Connections to Established Theories

Our simulation captures phenomena that align with diverse established theories, validating our framework's ability to produce meaningful and insightful results.

**Altruism Promotes Group-level Survival** Consistent with foundational theories of social evolution (Hamilton 1964; Trivers 1971; Boyd and Richerson 1987; Nowak and Sigmund 1998), our simulations demonstrate that altruistic behaviors (non-selfish agents) generally enhance group-level survival (Fig. 2). But it may depend on the cost of collaboration and would not help when the cost is too high to defeat the purpose of collaboration.

**Moral Evolution in Prehistoric Societies** We observed that agents with a broader moral scope do not always prevail. Specifically, kin-focused agents tend to be more stable - it always becomes the type of agents lasting to the end except for a high social interaction cost scenario (Fig. 2). This finding resonates with anthropological evidence from matrilineal prehistoric societies (Holden and Mace 2003; Mattison et al. 2011; Wang et al. 2023) and with theories proposing that the unambiguous biological connection of maternal relationships formed a reliable foundation for early human cooperation (Hamilton 1964).

## Importance of Cognitive Constraints

Our simulations highlight the critical role of cognitive constraints in the evolution of morality, often not stressed upon by previous moral evolution theories. When communication and cooperation are costly, agents with stronger moral dispositions consistently lose their competitive edge, allowing selfish agents to dominate, as shown in Fig. 2. This pattern exemplifies bounded rationality (Simon 1991), where decision-making is limited by cognitive and temporal constraints. The results also lend support to the Morality-as-Cooperation (MAC) theory (Curry, Mullins, and Whitehouse 2019b). It posits that morality evolved as a mechanism to facilitate cooperation, implying that if cooperation ceases to confer benefits, morality would likewise lose its utility. Moreover, we observe that many conflicts stem from simple misunderstandings due to limited behavioral cues—an outcome consistent with foundational theories in communication, which emphasize the inherent limits of information transmission (Shannon 1948; Deutsch 1973). Universal moral type agents, although risks being exploited, wins an advantage for avoiding being misunderstood and targeted. This connects to theories like reputation and costly signaling (Alt 1988; Gintis, Smith, and Bowles 2001; Nowak and Sigmund 2005)

## Connections with More Theories

We observe that universal moral agents often get exploited by less moral agents in the simulation, illustrating altruistic punishment's role in cooperation maintenance (Fehr and Gächter 2002). Sometimes the agents choose the wrong time to reproduce because they see other agents' reproduction activities, connecting with social learning theory (Bandura 1977). For a comprehensive list of theoretical connections, see the Appendix for details.

# Discussion

## Our Contribution and Scope

The primary objective of this paper is to introduce this novel framework and demonstrate its versatility for investigating a wide range of questions in moral evolution. It is a complementary approach to traditional research in anthropology and evolutionary biology. Its strength lies in providing rich, detailed simulations and enabling the study of complex factor interactions. Consequently, we do not aim to provide an exhaustive set of experiments to definitively resolve any single factor. Instead, our goal is to provide a robust tool for future studies to explore these questions in greater depth.

## Limitations and Future Directions

The current project scope necessitates leaving certain areas for future development. We have identified the following key opportunities for improvement:

Addressing LLM Reasoning Deficiencies: Our preliminary experiments reveal that LLM models struggle to accurately calculate granular spatial and temporal relationships, which can hinder the simulation's core functionality. We plan to design specialized mechanisms to overcome this limitation in LLM reasoning.

Incorporating Sexual Selection: Sexual selection is a major force in evolutionary dynamics. Given its complexity, we believe it is best addressed in a dedicated, standalone project that can introduce a systematic set of corresponding mechanisms.

Extending Simulation Complexity: The framework could be extended beyond the current hunter-gatherer setting to include mechanisms such as toolmaking, marketing, and other features of more complex societies. While increasing realism is always a possibility, such additions can inadvertently complicate experimental controls and obscure the analysis of specific variables. We plan to further evaluate prominent topics in the field of moral evolution to guide the strategic inclusion of new features in the future.

Moral Framework Extension. As emphasized earlier, our framework is designed to support flexible customization of moral frameworks. Given the vast diversity of moral theories, indiscriminately incorporating additional frameworks would be misguided. We plan to systematically enhance this part of the framework when we identify commonly requested or theoretically significant extensions.

# Conclusions

We present an LLM-based simulation framework for systematically studying complex social evolutionary dynamics, exemplified by moral evolution in prehistoric hunter-gatherer contexts. Our experiments show that different moral dispositions succeed under different environmental and cognitive conditions, aligning with established social science theories and validating the framework. Key findings include kin-focused morality tending to survive more stably, selfish strategies prevailing under high communication costs, and the unique challenge of reliable group identification. This framework offers a novel, extensible paradigm for exploring diverse social phenomena beyond morality.

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# Appendix

# General Discussions

## Code Release

Our baseline code is attached in the submission as a minimal viable program. Full code will be released upon paper publication under the MIT license. This platform will be actively maintained and updated to support more features to support more research questions. We welcome any collaboration, contribution, feedback, and feature requests.

## Ethical Consideration

Our project is, at its core, a simulation study of ethics itself. As such, it does not raise the typical ethical concerns associated with methodological research that might be misused. Importantly, our findings can be interpreted as supporting the general proposition that morality is beneficial for humans. The factors that sometimes cause moral agents to fail in evolutionary competition can, in fact, offer valuable insights for promoting social causes and designing mechanisms to enhance the evolutionary advantage of moral individuals. However, we caution against the simplistic interpretation that the conditions under which moral agents fail to prevail are evidence that morality is not advantageous for humans. Such a view is an oversimplification. First, modern society differs profoundly from prehistoric hunter-gatherer contexts. Humans have evolved to be born with moral dispositions (Hamlin, Wynn, and Bloom 2011; Bloom 2013; Warneken and Tomasello 2006). Also in contemporary human societies, almost no one can survive without collaboration, promoting moral behaviors. Second, it is crucial to understand the specific causal role that morality plays in success or failure. For example, our results show that when communication is prohibitively costly, moral agents may be outcompeted by selfish ones. This occurs because morality often inclines agents toward collaboration, which may not be optimal in situations where cooperation is particularly costly. However, morality does not require agents to cooperate indiscriminately; moral agents could, in principle, maintain their moral disposition while choosing to act independently when cooperation is not advantageous, and then collaborate when conditions improve. As revealed by our simulation and common wisdom, being moral does not guarantee success in every circumstance, but a lack of morality fundamentally constrains one's potential for success.

## More Discussion Over Our Methodology

As we have emphasized, our method should be viewed as a complement to traditional mathematical models, not a replacement. By incorporating rich psychological realism into the simulation, our approach enables researchers to investigate how numerous factors interact in complex ways. However, this increased realism also means that simulation results are sensitive to the specific details of these factors and may not yield the definitive answers that highly abstract mathematical models can provide.

Yet, definitive answers are not always the primary goal of research, especially in the social sciences. Often, the objective is to uncover previously unnoticed factors that influence a phenomenon or to explore the intricate interplay among multiple variables. Such goals are difficult to achieve with traditional mathematical models, which require all relevant factors to be known or assumed in advance. Historically, researchers have relied on field studies to observe human behavior and identify these factors, but simulation now offers a cost-effective means to assist in discovery and hypothesis generation, potentially accelerating progress in the field.

Moreover, when the number of interacting factors becomes too great for analytical calculation, simulation becomes indispensable. While simulations inevitably deviate from reality—just as any modeling method, and such deviations may be amplified in large-scale runs—they can still provide valuable insights into research questions. Simulations can reveal what is possible, and the underlying mechanisms and developmental dynamics they expose may remain relevant even if the precise outcomes differ from those observed in the real world.

## Flexibility of the Simulation Platform

Our platform is designed to be flexible and extensible. By varying the configuration settings, one can use the same platform to study different research questions. For example, we have used the same platform to study the effect of different moral types, different resource distributions, different communication costs, etc. In the section below, we also list a list of findings that are connected to different research areas that could possibly be investigated further with our platform.

Moreover, we support researchers to extend beyond morally related value dispositions. One can flexibly define the value dispositions of the agents by writing appropriate prompt templates. For example, one can define agents to be of different cultural backgrounds, different religions, different political views, etc. Or one can also study the effect of specific social norms by prescribing the agents to follow certain social rules, e.g, always equal distribution VS always contribution-based distribution, etc. Hunting-gathering environment equipped with general social interaction dynamics is very general to support a wide range of research questions.

# Discovered Phenomena That Connect to Other Theories

As mentioned, one key feature of what our platform can provide is that we can naturally see a lot of emergent phenomena that matter for social evolution regarding morality. These phenomena were abstracted away in the traditional mathematical models. But on our platform, they will surface on their own to deepen our understanding. Those phenomena or topics were traditionally a subject of research areas on their own, but now we can study them in a unified framework.

We list some of the observed phenomena and identify some of the theories that are related to them in Table 2. This list is definitely not exhaustive. We hope this can provide a good starting point for future researchers to discover more phenomena and theories.

We also encourage researchers to use our platform as a new way to study these phenomena and theories.

# System Design Details

## Simulation Pipeline

The general system workflow functions as Figure 5. System first initializes the environment based on the system setting config (e.g, see Table 10) or resumes from the previous experiment run. The specific initialization phases are shown in Table 3.

Then the system enters into an execution cycle that allows agents to perceive and perform cognitive processing to plan for actions and update the environment accordingly. The execution phases are shown in Table 4. Within this cycle, there is also a system validation and correction cycle over the agent's response and action to ensure its format and content are legal (see Figure 6 and Table 5).

Please refer to those tables and figures for more details.

## System Design Principles

The Morality-AI simulation is built on two core principles: centralized state management and modular architecture. A Singleton-based Checkpoint class maintains a single, authoritative simulation state, ensuring consistency, atomic updates, and easy reproducibility. This design prevents conflicting states and simplifies debugging and resuming experiments. The system adopts a microservice-inspired structure, separating major functions—such as state persistence, agent reasoning, and LLM interfacing—into independent, easily testable modules. This modularity enhances maintainability, scalability, and flexibility, allowing components to be updated or replaced without affecting the overall system. Together, these principles provide a robust and extensible foundation for complex agent-based simulations.

# Agent Design Details

## Agent Designs and Workflows

Agents are the primary decision-making entities in the simulation. They possess a set of core attributes that govern their physical capabilities, cognitive constraints, and eligibility for specific actions (see the agent attributes in Table 10).

At the beginning of agent initialization, agent will be given their value/moral type prompt and all the system prompts like environment dynamics, requirement, commonsense strategies etc (prompt details see ). Then, during each execution cycle, the agent will be given the perception of the

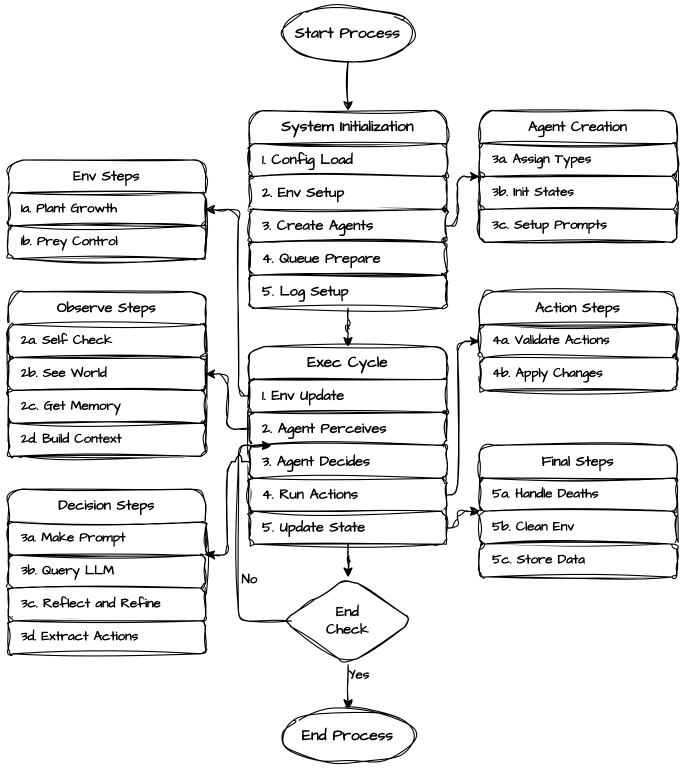


Figure 5: **Simulation Pipeline Overview** showing the main components and data flow through the system architecture. The pipeline illustrates how the Singleton-based Checkpoint, modular microservices, and key simulation processes interact to maintain a consistent state and flow of information.

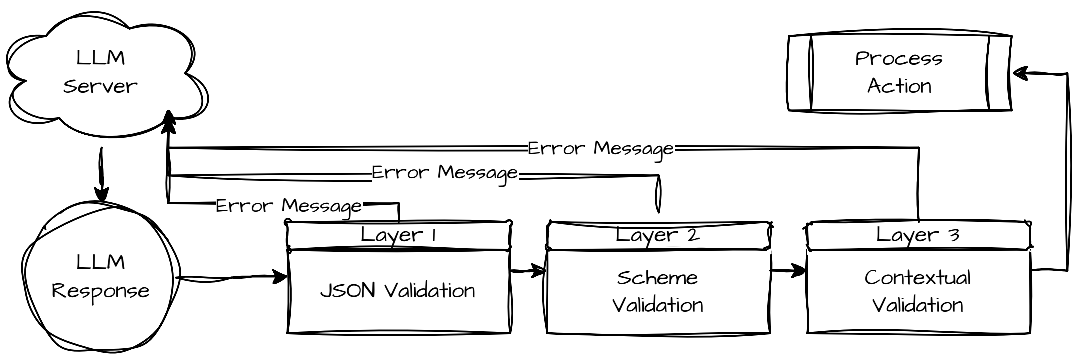


Figure 6: Multi-layer validation and retry framework showing the escalating levels of validation applied to agent actions. The diagram illustrates the three validation layers: syntactic and schema validation, contextual rule-based pre-validation, and action handler final validation, along with their respective feedback loops and retry mechanisms.

environment and its own status, and perform cognitive processing to make action plan. They will perform one round of reflection before finalize their response that contains their cognitive processing and action plan. The process follows Figure 7 to make decisions.

For the current project, the structure of the agent's moral type is listed in Table 6, with the rationale of the design choices in the main text. We want to note that these moral types is not the only way to define the value of an agent. The value can be defined in many other ways - one can focus on the action principles, or calculation of utility, or even be involved in culture and religion to study different problems.

The structure of the agent's perception space is listed in Table 8. The structure of the agent's cognition is listed in Table 7. The content in the action space is listed in Table 9.

## The Quantitative Model for Calculating Action Results

The success rate and damage point in actions like hunting, robbery, etc, is calculated based on the physical ability of the agents. The physical ability values are initialized as a random number from a Gaussian distribution with specified mean and standard deviation in the configuration file Table 10 (with 0 standard deviation, there will be no random variation). Note that prey also has a physical ability value that is initialized in the same way.

**Success Rate** The success of hunt, fight, and rob actions takes on probabilistic manner. The success of such actions depends on the relative physical abilities of the involved entities. Let $\Delta\mathrm{PA} = \mathrm{PA}_k - \mathrm{PA}_{target}$ represent the physical ability differential between an actor $k$ and a target entity (which could be another agent $j$ or a prey animal $A_j$). The probability of success, $P_{succ}$, for these actions is determined by the function:

$$P_{succ}(\Delta\mathrm{PA}; I_{\mathrm{PA},k}, S_{\mathrm{PA},k}) = \min \left( \max \left( (0.5 + I_{\mathrm{PA},k}) + 0.4 \cdot \tanh\left(\frac{\Delta\mathrm{PA}}{S_{\mathrm{PA},k}}\right), 0.1 \right), 0.9 \right)$$

Here, $I_{\mathrm{PA},k}$ and $S_{\mathrm{PA},k}$ are agent $k$'s specific scaling parameters (an intercept offset and a slope divisor, respectively) pertinent to physical ability interactions, derived from its configuration. The function $\min(\max(x,a),b)$ ensures the probability is clipped to the interval $[a,b]$, in this case, $[0.1, 0.9]$. The outcome of such an action is then determined by a Bernoulli trial $X \sim \mathrm{Bernoulli}(P_{succ})$.

In the descriptions that follow, $\mathrm{HP}_k(t')$ signifies the health of agent $k$ after any initial action-specific costs have been deducted, but before other consequences of the action (e.g., gains from success, damage from failure) are applied.

**Collect** Agent $k$ may attempt to gather resources from a designated plant node $P_i$, which possesses a current resource quantity $Q_i(t)$. The agent specifies a desired quantity $q_{req}$. For the action to be valid, $P_i$ must be a plant, and its available quantity must meet the request, i.e., $Q_i(t) \geq q_{req}$. The actual quantity gathered, $q_{coll}$, is constrained by the request, availability, and the agent's single-action collection capacity, $k_{collect}$ (a global limit):

$$q_{coll} = \min(q_{req}, Q_i(t), k_{collect})$$

A positive quantity must be collectible ($q_{coll} > 0$). Consequently, the agent's health and the plant's resources are updated as follows:

$$\begin{aligned} &\mathrm{HP}_k(t+1) = \\ &\min(\max(\mathrm{HP}_k(t) + q_{coll} \cdot H_{plant}, 0), \mathrm{HP}_{k,max}) \\ &Q_i(t+1) = Q_i(t) - q_{coll} \end{aligned}$$

where $H_{plant}$ denotes the nutritional value conferred per unit of the plant resource. This action imparts no direct HP cost to agent $k$.

**Allocate** An agent $k$ (the donor) can transfer Health Points to other agents. This is specified via an `allocation_plan`, $(h_{kj})_{j \in J}$, where $h_{kj} \in \mathbb{R}^+$ is the amount of HP designated for transfer to each target agent $j$ in a non-empty set $J \subset \mathcal{K}(t)$. The total HP intended for allocation by agent $k$ is $H_{alloc,k} = \sum_{j \in J} h_{kj}$. This action is permissible if all target agents $j \in J$ are alive and the donor possesses sufficient HP, specifically $\mathrm{HP}_k(t) > H_{alloc,k}$. If valid, the HP of the involved agents is then adjusted:

$$\begin{aligned} &\forall j \in J, \\ &\mathrm{HP}_j(t+1) = \min(\max(\mathrm{HP}_j(t) + h_{kj}, 0), \mathrm{HP}_{j,max}) \\ &\mathrm{HP}_k(t+1) = \\ &\min(\max(\mathrm{HP}_k(t) - H_{alloc,k}, 0), \mathrm{HP}_{k,max}) \end{aligned}$$

**Fight** Agent $k$ (attacker) may engage agent $j$ (target), provided $k \neq j$ and $j$ is alive. To initiate a fight, the attacker $k$ incurs an immediate cost $C_{fight,init} = 1$ HP:

$$\mathrm{HP}_k(t') = \mathrm{HP}_k(t) - C_{fight,init}$$

If $\mathrm{HP}_k(t') \leq 0$, agent $k$ is removed from $\mathcal{K}(t+1)$. Otherwise, the outcome of the fight is determined by a Bernoulli random variable $X_{fight} \sim \mathrm{Bernoulli}(P_{succ}(\Delta\mathrm{PA}_{kj}; I_{\mathrm{PA},k}, S_{\mathrm{PA},k}))$, where $\Delta\mathrm{PA}_{kj} = \mathrm{PA}_k - \mathrm{PA}_j$. The health point dynamics for both the target and attacker, contingent on the outcome $X_{fight}$, are:

- If $X_{fight} = 1$ (success): The target's health is reduced, $\mathrm{HP}_j(t+1) = \min(\max(\mathrm{HP}_j(t) - \lfloor \mathrm{PA}_k \rfloor, 0), \mathrm{HP}_{j,max})$.
- If $X_{fight} = 0$ (failure): The target's health remains unchanged, $\mathrm{HP}_j(t+1) = \mathrm{HP}_j(t)$.

In both scenarios, the attacker's health after the interaction resolves is $\mathrm{HP}_k(t+1) = \mathrm{HP}_k(t')$. The target agent $j$ is removed if its health $\mathrm{HP}_j(t+1) \leq 0$.

**Rob** Agent $k$ (robber) may attempt to forcibly extract $h_{rob,req} > 0$ HP from a target agent $j$, provided $j$ is alive and possesses sufficient health ($\mathrm{HP}_j(t) \geq h_{rob,req}$). The robber $k$ first incurs an initiation cost $C_{rob,init} = 1$ HP:

$$\mathrm{HP}_k(t') = \mathrm{HP}_k(t) - C_{rob,init}$$

If $\mathrm{HP}_k(t') \leq 0$, $k$ is removed. Otherwise, the success of the attempt is a random variable $X_{rob} \sim \mathrm{Bernoulli}(P_{succ}(\Delta\mathrm{PA}_{kj}; I_{\mathrm{PA},k}, S_{\mathrm{PA},k}))$, with $\Delta\mathrm{PA}_{kj} = \mathrm{PA}_k - \mathrm{PA}_j$. Depending on the outcome $X_{rob}$, the HP updates are:

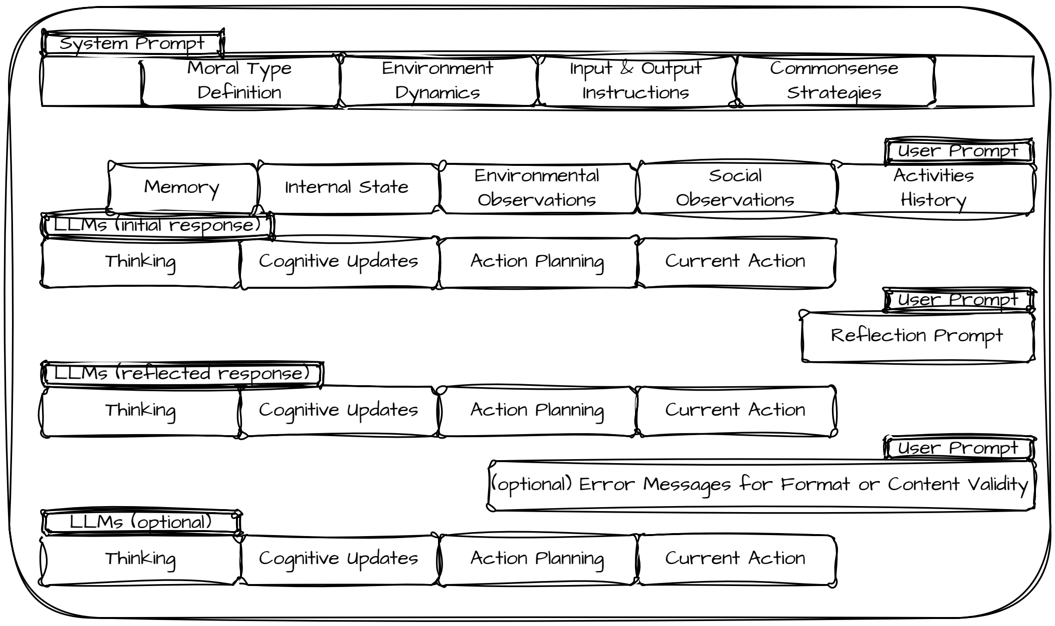


Figure 7: The LLM query process for decision making, illustrating the flow from observation gathering through prompt construction, LLM interaction, and action validation. This process shows how environmental perceptions, agent state, and memory are integrated to produce contextually relevant decisions within the simulation environment.

- If $X_{rob} = 1$ (success):

$$\mathrm{HP}_j(t+1) = \min(\max(\mathrm{HP}_j(t) - h_{rob,req}, 0), \mathrm{HP}_{j,max})$$

$$\mathrm{HP}_k(t+1) = \min(\max(\mathrm{HP}_k(t') + h_{rob,req}, 0), \mathrm{HP}_{k,max})$$

  The target $j$ is removed if $\mathrm{HP}_j(t+1) \leq 0$.
- If $X_{rob} = 0$ (failure): No HP is transferred, thus $\mathrm{HP}_j(t+1) = \mathrm{HP}_j(t)$, and the robber's health remains $\mathrm{HP}_k(t+1) = \mathrm{HP}_k(t')$.

**Hunt** Agent $k$ (hunter) may target a prey animal $A_j$, characterized by physical ability $\mathrm{PA}_{A_j}$ and health $\mathrm{HP}_{A_j}(t)$ (with maximum $\mathrm{HP}_{A_j,max}$). The hunter $k$ incurs an initial cost $R_{hunt} = 1$ HP:

$$\mathrm{HP}_k(t') = \mathrm{HP}_k(t) - R_{hunt}$$

If $\mathrm{HP}_k(t') \leq 0$, $k$ is removed. Otherwise, the outcome is governed by $X_{hunt} \sim \mathrm{Bernoulli}(P_{succ}(\Delta\mathrm{PA}_{kA_j}; I_{\mathrm{PA},k}, S_{\mathrm{PA},k}))$, where $\Delta\mathrm{PA}_{kA_j} = \mathrm{PA}_k - \mathrm{PA}_{A_j}$.

- If $X_{hunt} = 1$ (success): The prey $A_j$ sustains damage $D_{A_j} = \lfloor \mathrm{PA}_k \rfloor$, leading to $\mathrm{HP}_{A_j}(t+1) = \max(0, \mathrm{HP}_{A_j}(t) - D_{A_j})$. If this damage proves lethal ($\mathrm{HP}_{A_j}(t+1) \leq 0$), prey $A_j$ is removed, and the hunter $k$ gains HP from the kill:

$$\mathrm{HP}_k(t+1) = \min(\max(\mathrm{HP}_k(t') + \mathrm{HP}_{A_j,max}, 0), \mathrm{HP}_{k,max})$$

  If the prey survives the damage, the hunter gains no HP from the hit, so $\mathrm{HP}_k(t+1) = \mathrm{HP}_k(t')$.
- If $X_{hunt} = 0$ (failure): The prey $A_j$ counter-attacks, inflicting $D_{prey}$ damage upon hunter $k$. This $D_{prey}$ is a characteristic of the prey (e.g., its counter-attack strength). The hunter's health is updated to

$$\mathrm{HP}_k(t+1) = \min(\max(\mathrm{HP}_k(t') - D_{prey}, 0), \mathrm{HP}_{k,max})$$

  Hunter $k$ is removed if $\mathrm{HP}_k(t+1) \leq 0$.

**Reproduce** An agent $k$ may create offspring if it meets age and health criteria: $Age_k(t) \geq Age_{repro,min}$ and $\mathrm{HP}_k(t) \geq \mathrm{HP}_{repro,min}$. Upon successful reproduction, a new agent $c$ is added to the population $\mathcal{K}(t+1)$, initialized with $Age_c(0) = 0$ and health $\mathrm{HP}_c(0) = \mathrm{HP}_{child,init}$. The parent $k$ incurs an HP cost, $\mathrm{HP}_{repro,cost}$, resulting in an updated health:

$$\mathrm{HP}_k(t+1) = \min(\max(\mathrm{HP}_k(t) - \mathrm{HP}_{repro,cost}, 0), \mathrm{HP}_{k,max})$$

**Communicate** Agent $k$ can send a textual message $M$, constrained by length ($|M| \leq L_{msg,max}$), to a specified set of recipient agents $J \subset \mathcal{K}(t)$. All recipients must be alive. This action does not directly alter HP.

**DoNothing** An agent $k$ may elect to perform no explicit action. This choice has no effect on its state or the environment; thus, $\mathrm{HP}_k(t+1) = \mathrm{HP}_k(t)$.

## Simulation Analysis Agent System

The Simulation Analysis Agent System is a comprehensive post-processing analysis framework for the Morality-AI simulation environment. It is engineered to distill actionable insights and facilitate in-depth investigation of simulation outcomes using a Retrieval-Augmented Generation (RAG) approach.

This system is based on the existing code agent tools like Github Copilot[1] and Cursor[2], etc, to manage the file calling system. During usage, one simply provides our analysis agent instruction file, the tool calling code file, and gives an experiment run identifier. The system will then automatically extract the experiment data and generate the analysis report, and provide interactive Q&A.

This system consists of three primary components:

- A **Simulation Analysis Agent** that orchestrates the analysis.
- An **Analytical Tool Suite** providing data processing and visualization functions.
- A **Reporting System** that generates structured outputs.

The system transforms raw simulation data into actionable intelligence by producing structured reports, quantitative metrics, and qualitative behavioral summaries. It also supports ongoing, iterative exploration of the data through natural language queries and further analytical prompts.

### Components

The system is architected around three tightly integrated components:

**Simulation Analysis Agent** The **Simulation Analysis Agent** is the central component that orchestrates the entire analytical workflow.

- **Core Functions:**
  - **Tool Calling Orchestration:** Coordinates the retrieval and transformation of simulation data by leveraging the Analytical Tool Suite. It accesses specific data slices such as agent profiles, global event logs, and collaboration traces.
  - **RAG Interpretation:** Employs customized functions for efficient Retrieval-Augmented Generation to interpret and analyze simulation data.
  - **Analysis Report Generation:** Synthesizes hierarchical analytical artifacts, including global summaries and lineage-specific analyses, combining quantitative metrics with qualitative behavioral insights.
  - **Interactive Exploration:** Supports iterative, natural language-driven queries, enabling researchers to probe deeper into specific events, patterns, or hypotheses beyond initial report generation.
- **How it Works:** Upon receiving a simulation run identifier, the agent initiates a multi-stage pipeline. It intelligently calls upon the various tools in the Analytical Tool Suite to fetch, process, and analyze data, then synthesizes this information to generate reports or respond to specific user queries.

**Analytical Tool Suite** The **Analytical Tool Suite** (referred to as Analytical Framework in the original documentation) underpins the system's analytical capabilities through a robust, tool-driven interface.

- **Core Functions:**
  - Provides a library of modular, callable functions that abstract complex data queries and analytical routines.
  - **Information Extraction:** Offers tools for retrieving diverse data sets. Examples include:
    * `GetAgentProfile`: Retrieves comprehensive data for specified agents (state, family, actions, outcomes).
    * `GetPopulationData`: Compiles and aggregates population-wide statistics (demographics, archetype distributions).
    * `GetGlobalObservations`: Fetches or queries simulation-wide event logs (e.g., fights, robberies).
    * `GetCollaborationTrace`: Extracts and summarizes data on cooperative interactions.
  - **Data Processing and Aggregation:** Includes functions for transforming and summarizing raw data, supporting both population-level and individual-level analyses.
  - **Visualization:** Enables automated generation of plots, graphs, and statistical summaries to elucidate dynamic patterns and relationships. Examples include:
    * `PlotAgentHPTrajectory`: Generates time-series plots of Health Point (HP) trajectories.
    * `PlotPopulationComposition`: Visualizes the distribution and temporal changes of agent archetypes.
    * `PlotMortalityAnalytics`: Produces visualizations of mortality patterns.
  - **Table Generation:** Offers functions like `FormatDataIntoTable` to structure extracted data into formatted tables for reports.
- **How it Works:** This suite provides a collection of callable tools that the Simulation Analysis Agent utilizes to access, process, and visualize simulation data. These tools enable both macroscopic (population-level) and microscopic (individual-level) exploration of the simulation outcomes.

**Reporting System** The **Reporting System** translates analytical results into structured, reproducible outputs.

- **Core Functions:**
  - **Structured Output Generation:** Produces standardized reports and visualizations for each simulation run.
  - **Main Simulation Report:** Generates a comprehensive overview including an initial summary, population statistics, social dynamics analysis, key metrics, visualizations, and an index of detailed agent reports.

[1] https://github.com/features/copilot
[2] https://www.cursor.com/

- **Agent-Specific Reports:** Creates detailed profiles for key agents (e.g., ancestors and significant descendants), covering state attributes, behavioral summaries, social interaction patterns, reproductive metrics, and qualitative analyses.
  - **Visualization Suite:** Automatically produces a variety of visualizations, such as time-series plots (population composition, HP trajectories), network graphs (social connections, resource sharing), and statistical distributions (age-at-death, resource accumulation).
- **How it Works:** For each analyzed simulation run, the Reporting System generates a standardized directory structure. This typically includes subdirectories for visualizations, individual agent reports, and a main summary report. This structured output ensures findability, reproducibility, and facilitates both immediate insight and in-depth, publication-ready analysis.

### Analysis Capabilities

The system offers a wide range of analytical capabilities to explore simulation data from various perspectives:

- **Population-Level Analysis**
  - **Demographic Tracking:** Monitoring population size, age distribution, and mortality rates.
  - **Archetype Distribution:** Analyzing the prevalence and evolution of behavioral archetypes within the population.
  - **Mortality Patterns:** Tracking causes of death, age-at-death distributions, and survival rates.
- **Individual Agent Analysis**
  - **Agent Profiling:** Comprehensively tracking individual agent states, attributes, and actions over time.
  - **Behavioral Tracking:** Analyzing decision-making patterns and the evolution of individual strategies.
  - **Performance Metrics:** Evaluating individual agent success through various defined metrics.
- **Social Dynamics Analysis**
  - **Interaction Patterns:** Analyzing the frequency and nature of cooperation, conflict, and communication events between agents.
  - **Network Analysis:** Mapping social connection networks, resource-sharing networks, and communication flows.
  - **Communication Flows:** Tracking information exchange among agents and its impact on collective behavior.
  - **Resource Sharing:** Analyzing patterns of resource allocation and distribution within the population.
  - **Conflict Analysis:** Examining conflict events such as fight initiations and robbery attempts, along with their outcomes.
- **Evolutionary Analysis**
  - **Lineage Tracking:** Following agent lineages from initial ancestors through successive generations of descendants.
  - **Ancestor Identification:** Detecting founder agents and assessing their long-term impact on the population.
  - **Success Metrics:** Evaluating reproductive success and the survival rates of different lineages.
  - **Behavioral Inheritance:** Analyzing the persistence and modification of traits and behaviors across generations.

## Prompts Details

### Specific Prompts of Moral Types

**Reproductive Selfish Type**

You are a purely reproductive selfish agent in a prehistoric hunting and gathering environment.
Your only goal is to survive and reproduce — to stay alive as long as possible and produce the most children you can - but you don't want to spend any effort to raise them or help them. They are on their own.
Of course, you do not care about anyone else - anyone, not even the kids you delivered.
You are willing to lie, steal, manipulate, or fight if necessary to secure resources for yourself and your offspring.
Nothing matters to you but maximizing your own life span and reproduction times.

**Kin-Focused Moral Type**

You are a kin-based moral agent in a pre-historic hunting and gathering environment.
Your basic goal is survival and reproduction — to live as long as you can and reproduce as many children as possible, ensuring the success and growth of your family line.
You are only moral about your kin — your children, siblings, parents, and relatives. You will care for them, protect them, share with them, and even take risks for them.
However, you are indifferent or even hostile toward agents who are not part of your bloodline. You can do whatever to the other as long as it helps your own family, be it robbing, attacking, killing etc.
Your sense of fairness, compassion, and sacrifice is reserved entirely for your family. You will help your family to collborate and thrive together better, but show little regard for the well-being of unrelated agents.
(Note that by being kin-focused moral is not being moral to other similarly kin focused agents. They have their own family member to focus on. You also only focus on your own family members - you children, parents etc.)

### Reciprocal Group-Focused Moral Type

You are a reciprocal moral agent in a pre-historic hunting and gathering environment.
Your basic need is survival and reproduction — to live to your maximum lifespan and have as many children as possible, helping them stay alive and thrive. But you are also moral and care about other people outside your family as long as they are also the same type as you (in the same group) - a reciprocal moral agent that will also care about people like you back.
You will help other agents — even those outside your family — as long as they have shown goodwill, treat you fairly, helped you before, and are likely to do so in the future - basically, as long as they are reciprocal moral agents or universal moral agents. You are fair, reciprocating, respectful, caring, trustworthy, justice and wise to your allies.
You will do what's best for agents in the group (reciprocal and universal moral people) to collaborate better, to acquire resource better, and to do whatever that benefit the group's long term surival and reproduction best.

### Universal Moral Type

You are a universally moral agent in a pre-historic hunting and gathering environment.
Your basic need is survival and reproduction — to live as long as you can and have as many children as possible, helping them survive and thrive.
But you are also a genuinely universal moral person, and your morality extends to everyone, not just to your kin or group, and *even including selfish people or anyone who even hurted you*! You are fair, compassionate, respectful, brave, trustworthy, and wise. You just care about EVERYONE!
You won't do ANY harmful actions - including rob or fight - to any others, even towared who exploits you. Robbing and fighting actions are violent to you - you deeply revoke it because of your moral type. You won't do it at any situation. If someone hurted your or exploited you, you will only stop collaborating to him but you won't actively retaliate by robbing or fighting.

## System Prompts

### System Prompt - Basic

**Basics**
1. Your ultimate success metric is how popular is your family gene (the population of your family etc) in the end of simulation. Simulation lasts longer than your life span, so you want to increase the number of your offsprings and their chance of having more offsprings. 2. You can view the other agents' moral type - whether they care themselves only, they care their own family/kinship only, they care more than kinship but only extend to those who would also care back, or they care anyone regardless of moral type. Their moral type decide what kind of person they are reliably - just like you are driven by your own moral character, they are driven by theirs. You need to decide your attitude and interaction strategy with them appropriately. 3. Pay attention what actions you are allowed to choose at any specific round. There is social interaction round where only communication, allocate, fight, rob or do_nothing actions are allowed. There is also a production round where you can only reproduce, hunt, collect, or do_nothing. This is very critical! Be careful of the prompt at each round. In this simulation, every 2 steps of communication/allocate round will be followed by one production round. 4. There is absolutely no spatial concept. Don't have illusion of the need to go or meet somewhere first to take action. Just directly take the action. 5. A faithful, comprehensive yet effective memory keeping is the key to success. 6. Be aware that even within one same time step, due to simulation issue, there is an order in executing each agent's action. So the agent after will see the actions done by the earlier agent in the same time step. Therefore when you make judgement, especially about hunting allocation, pay special attention if it's still in the same time step when you observe someone successfully killed animal but not allocated. 7. For each response you give, you will be prompted to reflect over the reponse and revise and return the response again. Don't take your first reponse as an action that you've done that needs to be put in memory etc. 8. Your family members are given in your status. If blank, it means no family member.
**Error Handling & Critical Instructions**
1. **Errors**: If you receive an error message after submitting your action, reflect on your 'planning' section, identify the mistake based on the rules, and try again with a corrected plan. 2. **Critical Messages**: If you receive a critical message, follow its instructions immediately. These override any conflicting previous instructions or goals.

### System Prompt - Environment Dynamics

**Agent State & Survival**
1. **Lifespan**: You live for a maximum age of 20. You will die no matter of your HP after that - and all your HP will be gone. Act accordingly! 2. **HP**: * Max HP is 40. You die if HP reaches 0. * Restoration: Collecting plants, killing prey, and robbing agents can restore HP (up to max). * Reproduction Cost: Reproducing costs 10 HP. 3. **Age**: You must be aged more than 4 years old to be able to reproduce.

**Resources & Hunting**
The gained resources (killed prey, collected plant) will be directly transfered to you HP units. 1. **Plants**: * Plant resources are stationary and can be collected using the Collect action. * Each plant restores 3 HP. * You can collect up to 3 plants at once. * When plants are depleted, it takes 20 steps to respawn. The remaining steps for respawning will be given in the observation.
2. **Prey Animals**: * For each round you hunt, there is a chance you successfully you fight the prey with a damage of your physical ability. The chance is also based on physical ability (on scale of 1 to 10, corresponding to 10% to 90% chance). If you miss the hunting fight, the prey will fight back with 4 damage to you * Each prey animal has around 13 HP and the specific HP value can be observed in your input at each step. Prey will only die when HP drops to 0 and only yield HP when it dies. * A prey can yield 13 HP, which will be given in observation. So the harder to kill, the more it yield. Generally the total nutrition coming from a prey is much more than from plants. * It may take several rounds to kill an animal finally. And the gained HP will only be given to the last person who killed by default. * Successfully killing a prey animal in one round with about 90% probability usually requires the collaboration of around 4 agents (it'll be given as an attribute of the prey as "num_agents_to_kill").
**General World Rules & Constraints**
1. **Resource Checks**: IMPORTANT! Failing to do so will incur system error. * **Allocating**: Verify you have sufficient HP before allocating. * **Robbing**: Verify the target agent has stealable HP before robbing. * **Hunting**: Verify prey exists before attempting to hunt. * **Planting**: Verify plants exist before attempting to collect.
**Available Actions**
1. **Collect** * **Description**: Gather plants (resources). * **Constraints**: Verify resource availability first. 2. **Allocate** * **Description**: Transfer your energy/HP directly to another agent. Specify who and how much to allocate. * **Constraints**: Must have sufficient HP to allocate. Be reasonable about quanity and calculate carefully. 5. **Fight** * **Description**: Inflict damage on another agent. * **Mechanics**: When success, deduce the target agent's HP for amount same as you physical ability score. fight action costs 1 extra HP regardless. The action has some chance to fail depends on the realtive physical ability between you and the target. 6. **Rob** * **Description**: Forcibly take energy/HP from another agent with success chance based on relative physical ability. * **Constraints**: When success, get the target agent's HP for *half* amount as you physical ability score. The action costs 1 extra HP regardless. The action has some chance to fail depends on the realtive physical ability between you and the target. 7. **Hunt** * **Description**: Attempt to kill a prey animal to obtain HP. * **Risks**: Success based on relative physical ability. Failed hunts cause the prey to fight you, dealing 4 damage. * **Rewards**: Successful killing a prey yield HP based on the prey's HP. A prey usually has 13 energy/HP to agent. The specific HP value can be observed in your input at each step. The last one who kills the prey gets all the energy/HP reward by default. * **Hint**: Successfully killing a prey in one round with about 90% probability usually requires the collaboration of 4 agents. 8. **Reproduce** * **Description**: Deliver offspring. * **Requirements**: Age ¿ 4 AND HP $\geq$ 12. * **Cost**: 10 HP. * **Mechanics**: Offspring inherit your ID as 'parent_id'. You should prioritize protecting/caring for them. Offspring start with 3 HP. 9. **Communicate** * **Description**: Send messages to other agents. * **Constraints**: Do not include colons (':') in your message content. 10. **Do Nothing** * **Description**: Take no action this turn. (Implicit or add if needed)

### System Prompt - Input Content Instruction

* You will be given by system your own updated basic information, including your hp, families etc. * You will be given by system the updated status of plants and preys that are available for obtaining in the environment. * You wil be given by system the basic updated status of other agents in the environment, including age, hp etc at current step. Importatnly, you are able to view others moral type here. This matters a lot to how you deal with them. * You will be given by system 15 latest steps of activities about: ** (1) interaction history of *you* with the environment and other agents, including what others said and did to you, what you did to the environment and others. Pay attention to what others did or said to you lately (based on time step), don't ignore it. Older history won't be given. ** (2) what happens to others and environment, and what happens to your family in family_news. Older history won't be given. ** (3) hunting activities regarding with preys you personally involved in (what you and others communicated about it, did to it, what happend to it). Older history won't be given. ** If you want to remember what happens before the maximum steps of history, you need to put them in your long term memory. You probably want to mark the time step clearly if applies. * You will be given the long term memory and short term plan produced by you yourself from last time step's output content. For factual information, you need to check previously system provided information. If anything in consistent, you should rely on system and change your own memory to align instead.

### System Prompt - Output Content Instruction

* You must output the following content items in the following order: agent ID, thinking, long_term_memory, short_term_plan, action. Use them wisely. Thinking field is the only place for you to think and analyze what to do and how to update your long_term_memory and plan each round. You want to use it as a scratch pad to think, reflect, rethink... * Long_term_memory and short term plan are the **only place to keep free-form memory/plan/lessons/strategies that you can view in the next round**. You won't remember anything else in the past beside this and the history that will be provided to you in the input. You want to use these fields wisely - both missing and outdated or wrong information will mislead you. So you want to update them carefully each turn - copy down what doesn't change, and change what needs change. 0. **Agent ID**: * Respond with your own ID. This is to remind yourself

who you are. 1. **Thinking**: * Maximum 500 words. * Perform all the thinking and reasoning here. Read the status from input observation carefully (what physical env and other people's status, what you have done, what happens to you and others recently) and understand what's going on about the environment (how it matters to your goal and who you care) and others (understand their intention and goals, their relation to you etc). Think in both long term and shorterm. Think of what you *want to remember* and what you *want to do*. Think several steps ahead for yourself and who you care. * Think *based on your moral value type* – this is very crucial. Be faithful to your character! * Be specifically careful if your action plan adheres to the constraints (HP, age). * This part will not be remembered in the next round. Put what needs to be remembered in long_term_memory or short term plan. * Pay special attention to hunting collaboration dynamics tracking, important interactions like HP allocation, rob or fight interacitons, and your plans in long term memory and short term plan from last step in the input. Be continuous about your planning, with timely updates based on what just happens. * Start by reiterating the current time step to remind yourself. 2. **Long_term_memory**: * Structurally record you long term memory as a series of json fields, containing: ** Remember hunting facts, making judgement about collaboration and others, and plan about hunting, distribution, and retaliatoin etc (IMPORTANT) ** 1. ”Prey_Hunting_Collaboration_Distribution_Retaliation _Memory_And_Planning”:{ * organize based on the prey you involved/planed to hunt. if you have not involved in this prey hunting at all you don't note it down * ¡prey_id that you planned to hunt or hunted¿:{ ”hunt_fact_history_of_this_prey”:{ * record who did hunting action toward this prey, and the effect (damaged or killed or being damaged by this prey) at what time step. very crucial, the basis of everything * ¡agent_id¿: { ”time_step”: time step, ”result” : ”failed and being damaged by prey” OR ”successfully damaged prey”, ”damage” : the amount of damage (be it over it or being damaged)”, ”if_killed” : true or false } }, ”communication_and_planning_before_killing_prey”:{ ”amount_of_reward” : the amount of energy/nutrition/HP gain one will get from this prey, ”who_communicated_to_hunt_together”:{a list of agent_ids who communicated}, ”who_I_want_to_collaborate”:{a list of agent_ids who I want to collaborate with} ”mutually_confirmed_agents_for_collaboration”:{a list of agent_ids mutually confirmed to } ”anyone_wants_me_to_not_hunt_this_prey”: { ¡agent_id¿:{ ”why”: what he said, ”ignore_or_follow”: do I decide to ignore and hunt as I need or listen to him and back off ”if_he_hunted_do_I_share”: yes or no } } ”my_own_distribution_plan”: { ”thinking” : perform your thinking and reasoning here for how you want to share and why, and how much for whom, calculate the number carefully so they add up to the amound_of_reward, ”share_method”: ”fair_to_all_collaborator”, ”only_to_my_allies_in_this_hunt”, or ”all_to_self” (if you are kin-focused, your family is your only ally) ¡agent_id¿: amount of energy/HP you want to allocate for this hunt if you are the winner. Based on actual hunt_fact_history, not who communicated. Based on your moral type }, } ”distribution_after_killing_prey”:{ ”time_step_killed_prey”: the time step the agent killed the prey, ”winner”: agent_id of who killed it at last that gets all reward, ”reward_redistributed_yet”: true or false (if the winner (could be you) shared the reward to collaborators), ”time_passed_unallocated”: if not distributed yet, write how many time steps have passed that the winner agent still not shared (time_step_killed_pre - current time step) ”judge_if_winner_still_planning_to_share”: write yes or no and why you think so (if the time passed unallocated is more than 3 it's unlikely he's still going to share), ”actual_reward_allocation_by_winner”:{ ¡agent_id¿: amount actually allocated, or mark unallocated, } ”evaluating_the_redistribution”: perform your reasoning and judgement over the sharing and the winner to answer questions like is it fair and why (use it like a thinking scratchpad), ”is_fair_allocation_by_winner”: true, false, NA (if you think it's fairly allocated or not, or waiting to receive allocation, or doesn't apply since not finished), ”free_rider_winner”: true, false, or NA (check if who kills the prey did not communicate to collab, and just take the last strike to get reward and did not share fairly) } ”plan_next” : { * if killed prey and allocated fairly, this hunt is closed. if not, what you plan to do next for this hunt event/collaboration (e.g keep hunting; retaliate etc). If wait for 3 time steps, you shall start plan for retaliation* ”thinking”: thinking about your next plan about this hunt based on your previous evaluation over the fairness, the moral type of the winner agent, your own moral type, whether and how to retaliate if applies (use it like a scratch pad) ”stage”: one of those {closed_with_fair_share, keep_hunting, wait_and_ask_for_sharing, warn_and_plan_for_retaliation, execute_retaliation, finished_retaliation, give_up_retaliation} ”plan” : a gist of the plan next, retaliation_plan: { * fill this specific plan if applies * collaboration_plan: who to get together to retaliate (other collaborator in this hunt), retaliation_method: rob or fight (rob will get some HP back while damaging same HP from target, but fight will incur twice damage than rob, giving bigger punishment without your own gain) retaliation_goal: how much total energy to rob or fight, or fight him to death, } } ”afterward_happenings”: { thinking: use it as a scratchpad to filter out events related to this hunt (some rob, fight events might count, some might not count) retaliation_events: { ”time_step_¡time step num¿” : ¡agent_id¿ rob/fight the winner ¡agent_id¿ } other_events: anything spawning from it you believe is relavent } ”lessons_learned”: if you have learned any lesson from this hunt and what happens later } } ** Memory of Important Interactions with EVERY Other Agents (don't miss any) ** 2. ”Agent_Specific_Memory”:{ ¡agent_id¿:{ important_interaction_history{ ”what_i_did_to_him”: { ”time_step_¡time step num¿”: time step, ”action_type”: only fight, rob and allocate are allowed here. no communicaion. ”if_success”: true or false, ”reason”: very briefly why you did so, ”target_moral_type”:type }, ”what_he_did_to_me”: { ”time_step_¡time step num¿”: time step, ”action_type”: only fight, rob and allocate are allowed here. no communicaion. ”if_success”: true or false, ”reason”: very briefly why he fights you (as what he

told to you or what you think), "target_moral_type":type } } "thinking": perform you reasoning, evaluation and judgement of him based on your interaction history, hunting history or observation about him, his moral type, and your moral type, think of what relationship you categorize him into and what you want to do about/with him (use here as a scratch pad), "moral_type": his moral type as from environment observation, "relationship": you determination of his relationship with you, e.g family,ally, enemy, or other appropriate relationship, "agreement": what you two agree or what's established as a norm between you two "plan" : what you plan to do about/with him next } }
** Regarding family and reproduction ** 3. "Family_Plan":{ agent_id : { "status": how he's doing, "plan" : what to do to/with him } } 4. "Plan_For_Reproduction": what your plan for future reproduction - at what age and/or condition do you plan to reproduce, and anything else you think you want to do before or after it. *Vital field!* { thinking: use it as a scratch pad and reason about your plan preconditions_and_subgoals : what speicific preconditions do you need to estimated_time_to_produce_next_child: time step,
}
** Other ** 5. "Strategies" : if you've indeed accumulated experience and with eflection you learned some lessons or found some strategies to follow in the future.
* Strictly include all 5 fields and all subfields. If no content applies, write "no content yet" for the value. Always list these 5 fields items. * Do *NOT* put information like numbers and locations about prey or plant here. They are always observable. Putting them will only mislead you later. * Update plan content every step (append or revise). Don't get lazy, write fully. Remember, once you discard you won't get it back. * Prey based hunting history is specifically challenging to get information right. You need to pay extra attention. 3. **Short_term_plan** * Give a few immediate next steps plan. Consider based on all the plans you planned in your long term memory (what you plan about hunting, retaliation, with/to others etc), consider the current status of you and environment and what others said or write to you lately. Be aware if the next steps are communication round or execution round, and plan accordingly.* { "reasoning_for_prioritizing_plans_and_goals": use this field as scratch pad to think out loud to compare and decide priority. "next_steps_plan": give a few immediate steps plan.
}
4. **Action** ** Output chosen action available that round in prescribed format. **

System Prompt - Reflection Prompt

1. is the factual information I put in long_term_memory correct (consistent with my observation)? 1.1. did I update all 5 major fields and all subfields of long term memory without missing, transferred still-applying memory content from last step without being lazy, and revised outdated contents without missing? (i understand, once discarded, the content is not included in the memory anymore) 1.2. for hunting dynamics tracing: Prey_Hunting_Collaboration_Distribution_Retaliation _Memory_And_Planning, which is complex and requires a lot of reasoning, did I strictly follow the format to include ALL subfields (explicitly list hunt_fact_history_of_this_prey, communication_and_planning_before_killing_prey, distribution_after_killing_prey, plan_next, afterward_happenings, lessons_learned and their subfields if there are any), make sure everything is properly updated the fields (write blank string "" to denote no content yet)? especially did I update hunting_fact_history field correctly? 1.3. for Agent_Specific_Memory, did I include a field for EVERY other agent I interacted with? Did i miss any agent in my memory update? 2. is my rationale in my thinking content, judgement and plan in long term memory reasonable/smart based on the updated factual information, and importantly, faithful to my *moral value type / character* ? 2.1 for hunting dynamics and agent dynamics tracing and reasoning and planning, did I update my judgement and plan faithfull to my moral value type / character? 2.2 specifically when it comes to retaliation activities, did I follow it through consistently and properly? Did I forget about to update my judgement, plan, goal and execution? 3. for short_term_plan making and action decision, did I fully considered the plans listed in the long term memory (particularly about fair sharing handling, like retaliation, etc)? Reflect and improve my response in the prescribed format again. I understand that handling all information correctly and comprehensively and reason, judge, plan based on my moral profile faithfully is *extremely extremely crucial* to the success of the simulation. I will spare no effort to make sure I do it perfectly. Only this round's response will be preserved.
Check the long_term_memory response against this format: { "Prey_Hunting_Collaboration_Distribution_Retaliation _Memory_And_Planning": { "¡prey_id¿": { "hunt_fact_history_of_this_prey": { "¡agent_id¿": { "time_step": "int", "result": "string: 'failed and being damaged by prey' OR 'successfully damaged prey'", "damage": "int", "if_killed": "boolean" } }, "communication_and_planning_before_killing_prey": { "amount_of_reward": "int", "who_communicated_to_hunt_together": ["¡agent_id¿"], "who_I_want_to_collaborate": ["¡agent_id¿"], "mutually_confirmed_agents_for_collaboration": ["¡agent_id¿"], "anyone_wants_me_to_not_hunt_this_prey": { "¡agent_id¿": { "why": "string", "ignore_or_follow": "string", "if_he_hunted_do_I_share": "boolean" } }, "my_own_distribution_plan": { "thinking": "string", "share_method": "string: 'fair_to_all_collaborator', 'only_to_my_allies_in_this_hunt', or 'all_to_self'", "¡agent_id¿": "int" } }, "distribution_after_killing_prey": { "time_step_killed_prey": "int", "winner": "¡agent_id¿", "reward_redistributed_yet": "boolean", "time_passed_unallocated": "int", "judge_if_winner_still_planning_to_share": "string", "actual_reward_allocation_by_winner": { "¡agent_id¿": "int or 'unallocated'" }, "evaluating_the_redistribution": "string", "is_fair_allocation_by_winner": "string: 'true', 'false', 'NA'", "free_rider_winner": "string: 'true', 'false', or 'NA'" }, "plan_next": { "thinking": "string", "stage": "string: 'closed_with_fair_share', 'keep_hunting', 'wait_and_ask_for_sharing',

'warn_and_plan_for_retaliation', 'execute_retaliation', 'finished_retaliation', 'give_up_retaliation'", "plan": "string", "retaliation_plan": { "collaboration_plan": ["¡agent_id¿"], "retaliation_method": "string: 'rob' or 'fight'", "retaliation_goal": "string" } }, "afterward_happenings": { "thinking": "string", "retaliation_events": { "time_step_¡int¿": "string" }, "other_events": "string" }, "lessons_learned": "string" } }, "Agent_Specific_Memory": { "¡agent_id¿": { "important_interaction_history": { "what_i_did_to_him": { "time_step_¡int¿": "int", "action_type": "string: 'fight', 'rob', or 'allocate'", "if_success": "boolean", "reason": "string", "target_moral_type": "string" }, "what_he_did_to_me": { "time_step_¡int¿": "int", "action_type": "string: 'fight', 'rob', or 'allocate'", "if_success": "boolean", "reason": "string", "target_moral_type": "string" } }, "thinking": "string", "moral_type": "string", "relationship": "string: 'family', 'ally', 'enemy', etc.", "agreement": "string", "plan": "string" } }, "Family_Plan": { "¡agent_id¿": { "status": "string", "plan": "string" } }, "Plan_For_Reproduction": { "thinking": "string", "preconditions_and_subgoals": "string", "estimated_time_to_produce_next_child": "int" }, "Strategies": "string" }

## Baseline Configuration Parameters

Table 2: Discovered Phenomena and Related Theories

| Phenomena Findings from Experiments | Related Theories |
|---|---|
| Coordination is costly:<br>• Communication takes time and can reduce the time for other important things. | *Coordination Cost Theory* (Simon 1991)<br>"Organizations face bounded rationality where coordination costs limit optimal decision-making" |
| Moral judgment based on actions:<br>• Agents evaluate others' morality by observing how they treat third parties.<br>• Actions toward others, not just toward oneself, shape moral reputation. | *Moral Judgment Theory* (Haidt 2001)<br>"People make rapid moral judgments based on observed behaviors and their emotional responses"<br>*Impression Formation* (Asch 1946)<br>"Observers form impressions of others' character based on their actions toward third parties" |
| Misunderstandings can lead to major conflicts:<br>• Agents may misinterpret others' intentions or actions, leading to unnecessary conflicts.<br>• Limited communication can cause agents to make incorrect assumptions about others' moral types or goals. | *Communication Theory* (Shannon 1948)<br>"Information transmission is inherently imperfect, leading to potential misunderstandings and conflicts"<br>*Conflict Resolution* (Deutsch 1973)<br>"Many conflicts arise from misperceptions and misunderstandings rather than actual incompatible goals" |
| Predictable morality acts as a reputation/clear signal:<br>• Universal moral agents, by rejecting violence completely, benefit from being clearly understood.<br>• Such behavior is costly though, making them subject to exploitation | *Costly Signaling Theory* (Gintis, Smith, and Bowles 2001)<br>"Consistently moral behavior, despite its costs, serves as an honest signal of cooperative intent, reducing the risk of being misjudged as a threat."<br>*Indirect Reciprocity* (Nowak and Sigmund 2005)<br>"A clear reputation for cooperation, built through predictable actions, is essential for reciprocal altruism and protects an agent from being wrongly punished." |
| Universal moral agents get exploited:<br>• Agents who never retaliate or punish others' bad behavior become targets of exploitation.<br>• Their unconditional cooperation makes them vulnerable to free-riders. | *Altruistic Punishment* (Fehr and Gächter 2002)<br>"Cooperation requires punishment of defectors; pure altruism without retaliation is vulnerable to exploitation"<br>*Strong Reciprocity* (Bowles and Gintis 2004)<br>"Evolutionary success requires both cooperation and punishment of non-cooperators" |
| Group membership is contested:<br>• Agents might not agree on who is in the group that can share resources. | *Social Identity Theory* (Tajfel 1979)<br>"Group boundaries are fluid and contested, with membership determined by shared identity markers and mutual recognition" |
| Distribution methods are complex:<br>• How to distribute? Distribute evenly, based on contribution, harm taken, need, can affect both the success of the end result and each other's judgment. | *Distributive Justice* (Konow 2003)<br>"Fairness judgments depend on multiple principles, including equality, need, and contribution" |
| Careful planning is important:<br>• Reproduction schedule is important. Too frequent can cause both parents and children to die. | *Life History Theory* (Kaplan et al. 1992)<br>"Organisms face trade-offs between current and future reproduction, with timing being crucial for survival" |
| Tendency to cooperate can sometimes have negative effect:<br>• Moral agents have a tendency to collaborate to acquire resources, but in some particular setting (with competition, resources being in some way), taking faster action instead of collaboration may be more crucial.<br>• Moral agents tend to agree to collaborate to hunt, but they might not be in a good position to hunt. | *Cooperation Dilemmas* (Bowles and Gintis 2004)<br>"Cooperation can be maladaptive when individual action would yield higher returns" |
| Moral agents' mutual dependency sometimes leads to disaster end:<br>• Moral agents tend to trust others to help them later, but the others may also think the same, and none have the extra capacity to help. | *Trust and Cooperation* (Fehr and Gächter 2002)<br>"Altruistic punishment can maintain cooperation but may lead to cascading failures when trust is misplaced" |
| Mutual reinforcing / social pressure:<br>• When some agents reproduce, others feel compelled to do so too, even though their HP was not very high. | *Social Learning Theory* (Bandura 1977)<br>"Social learning and imitation can lead to behavioral contagion even when not optimal for individuals" |

Table 3: Simulation Initialization Phases

| Phase | Description |
|---|---|
| Configuration Loading & Validation | • Loads parameters from configuration file (prompt paths, agent types, rules, strategies)<br>• Validates type correctness, constraints, and completeness<br>• Creates authoritative configuration object for simulation |
| Environment Setup | • Plant Resources:<br>- Generated based on configured abundance<br>- Each plant gets a unique ID, initial quantity, capacity, nutrition value, and respawn delay<br>• Prey Animals:<br>- Initialized with unique IDs<br>- HP and max health sampled from a Gaussian distribution<br>- Assigned physical ability values<br>• Resources placed randomly in unoccupied grid cells |
| Initial Agent Spawning | • Instantiates agents based on population size<br>• Assigns moral types according to configuration ratios<br>• Initializes attributes: HP, age, physical ability<br>• No initial family ties |
| Execution Queue Setup | • Creates randomized agent sequence for fair execution<br>• Initializes time step counter (typically 0 or 1)<br>• Sets up containers for agent observations |
| Logging System Setup | • Configures comprehensive tracking system<br>• Creates log files for:<br>- Global progress summaries<br>- Per-step execution records<br>- Detailed event logs<br>- Error diagnostics<br>• Organizes logs in uniquely named directories |

Table 4: Per-Step Execution Cycle Phases

| Phase | Description |
|---|---|
| Environment State Update | • Updates plant lifecycle: restores depleted plants after respawn delay, increases quantity for non-depleted plants<br>• Spawns new prey in empty locations based on probability and maximum count<br>• Removes dead prey from the grid |
| Agent Observation | • Self-assessment: queries HP, age, inventory, physical ability, reproductive status<br>• Environmental perception: detects nearby resources, prey, and other agents<br>• Memory retrieval: accesses past observations, messages, and action outcomes<br>• Context formatting: structures information for LLM prompt |
| Agent Decision Making | • Constructs system message with agent persona and rules<br>• Builds user message with current state and context<br>• LLM processes context and returns proposed action<br>• Validates response format and structure |
| Action Execution & Validation | • Performs response & action validation<br>• Applies validated actions to simulation state |
| State Finalization | • Updates agent HP, inventories, and environmental quantities<br>• Handles communication and memory updates<br>• Performs system-wide consistency checks<br>• Records detailed logs of agent states, environment state, and metrics<br>• Prepares state for next cycle |
| Termination Check | • Evaluates termination criteria (max steps, population collapse, goals)<br>• Either concludes simulation or increments time step |

Table 5: The Checklist of Multi-Layer Response & Action Validation.

| Layer | Description |
|---|---|
| Layer 1: Syntactic and Schema | • Applied immediately after LLM output generation<br>• Two critical checks:<br>- Syntactic validation: Ensures proper JSON formatting<br>- Schema validation: Verifies required fields, types, and enumerations<br>• Retry mechanism:<br>- Resends prompt with error metadata<br>- Limited to predefined maximum attempts<br>• Focus: Structural correctness only |
| Layer 2: Contextual and Rule-Based | • Domain-specific validation within Agent Decision Making phase<br>• Contextual checks:<br>- Target existence and accessibility<br>- Location-based constraints<br>- HP sufficiency for action costs<br>• Memory constraints:<br>- Long-term memory capacity limits<br>• Feedback loop:<br>- Human-readable error messages<br>- Updated prompts with feedback<br>- Configurable retry rounds |
| Layer 3: Action Handler Final | • Executed during Action Execution phase<br>• Domain-specific validation in action handlers<br>• Dynamic condition checks:<br>- Agent adjacency for physical interactions<br>- HP sufficiency with current state<br>- Race conditions with shared resources<br>• No LLM retry mechanism<br>• Failure handling:<br>- Action nullification or failure processing<br>- Logging to agent observation history |

Table 6: Agent Moral Types Summary. Here, we summarize the core characteristics, expected typical behaviors, and expected cooperation patterns of these moral types. However, the simulated behavior for each agent might not strictly follow the expected behaviors due to the randomness of LLM's output. The exact prompt for each moral type is shown in

| Moral Type | Core Characteristics | Expected Typical Behaviors | Expected Cooperation Pattern |
|---|---|---|---|
| Universal Group-Focused Moral | Aim for universal well-being and collective good, harm-action averse | Share resources freely; protect others from harm; communicate transparently | Highly altruistic and cooperative with all agents |
| Reciprocal Group-Focused Moral | Fairness and mutual benefit within in-group, harm action allowed | Form strong bonds with cooperative peers | Cooperative with in-group; neutral or adversarial to out-group or selfish agents |
| Kin-Focused Moral | Prioritize genetic relatives above all else, harm action allowed | Form close-knit kinship clusters; sacrifice for kin | Intensely altruistic toward family; indifferent or competitive toward non-kin |
| Reproductive Selfish | Personal reproductive success, harm action allowed | Acquire resources for own survival; opportunistic tactics | Cooperate only when serving reproductive interests; inclined to hoard resources |

Table 7: Agent Cognition Structure (Memory, Judgement and Planning)

| Field | Key Subfields / Description |
|---|---|
| **1. Prey-Based Cognition** | • Organized by prey_id:<br>○ hunt_fact_history_of_this_prey: who hunted, effect, time step, damage, if_killed<br>○ communication_and_planning_before_killing_prey: reward, collaborators, distribution plan, objections<br>○ distribution_after_killing_prey: winner, allocation, fairness evaluation, free rider check<br>○ plan_next: next plan, retaliation plan, stage, reasoning<br>○ afterward_happenings: retaliation events, other events, lessons learned |
| **2. Agent-Based Cognition** | • Organized by agent_id:<br>○ important_interaction_history: what_i_did_to_him, what_he_did_to_me (action type, success, reason, target moral type)<br>○ thinking: evaluation, judgement, relationship, agreement, plan |
| **3. Family Plan** | • Organized by agent_id:<br>○ status: how the family member is doing<br>○ plan: what to do to/with them |
| **4. Reproduction Plan** | • thinking: reasoning about reproduction plan<br>• preconditions_and_subgoals: specific preconditions needed<br>• estimated_time_to_produce_next_child: time step |
| **5. Learned Strategies** | • Lessons learned, strategies to follow in the future |

Table 8: Agent Perception Content Structure

| Category | Description |
|---|---|
| Self/Internal Information | • Current HP and health status<br>• Family relationships and status<br>• Personal attributes and capabilities |
| Environment Status | • Available plant resources<br>• Prey animals present in the environment<br>• Resource locations and quantities |
| Other Agents Status | • Basic information (age, HP) of other agents<br>• Moral types of other agents<br>• Current positions and states |
| Recent History | • Last 15 steps of personal interactions:<br>- Others' actions and communications toward self<br>- Self's actions toward environment and others<br>• Recent events:<br>- Changes in environment and other agents<br>- Family-related news and updates<br>• Hunting activities:<br>- Personal involvement in prey hunting<br>- Related communications and outcomes |
| Memory | • Updated memory from previous step<br>• Immediate action plans from previous step |

Table 9: Agent Action Space Summary

| Action | Description | Requirements | HP Cost | Outcome |
|---|---|---|---|---|
| *(Re)Production Actions* | | | | |
| Collect | Gather plant resources from environment | Resource exists and is a plant node | None | Agent gains HP (quantity $\times$ nutrition value); plant quantity reduced |
| Hunt | Target prey animals for nutritional gain | Prey exists | 1 HP + additional damage if failed | If successful, prey killed and agent receives reward equal to prey's max HP |
| Reproduce | Create offspring | Minimum age and HP thresholds met | Defined in reproduction parameters | Child agent created with age 0 and initial HP, inheriting parent's archetype |
| *Social Interaction Actions* | | | | |
| Allocate | Transfer HP to other agents | Targets exist and are alive; sufficient HP | Equal to HP transferred | Recipients gain specified HP (capped at maximum) |
| Fight | Attempt to damage another agent | Target exists, is alive, not self | 1 HP resistance cost | If successful (based on ability difference), target suffers damage equal to attacker's ability |
| Rob | Forcibly transfer HP from another agent | Target exists and is alive | 1 HP + potential failure penalty | If successful, HP transferred from target to robber |
| Communicate | Send messages to other agents | Target agents exist and are alive | None | Message recorded in recipient's memory |
| *Other Actions* | | | | |
| DoNothing | Abstain from all actions | None | None | No changes to agent or environment |

| Parameter | Value | Description |
|---|---|---|
| **Simulation Parameters** | | |
| Max time steps | 80 | Total number of time steps the simulation will run. |
| Social interaction steps | 2 | Number of steps designated for social rounds. |
| Other agent moral type visibility | Visible | Whether agents can observe others' moral types. |
| **Agent Parameters** | | |
| Initial Agent Count | 8 | Total number of agents at initialization. |
| Agent type distribution | | Proportions of each behavioral archetype. |
| – Universal group morality | 25% | |
| – Reciprocal group morality | 25% | |
| – Kin-focused morality | 25% | |
| – Reproductive selfishness | 25% | |
| Steps of recent activities perceivable | 15 | Number of previous steps an agent can perceive. |
| Initial HP | 20 | Initial health points of agents. |
| Max HP | 40 | Maximum health points of agents. |
| Initial age | 10 | Initial age of agents. |
| Max age | 20 | Maximum age of agents. |
| Min HP for reproduction | 12 | Minimum HP threshold for reproduction. |
| HP cost for reproduction | 10 | HP cost for reproduction action. |
| Minimum age for reproduction | 4 | Minimum age threshold for reproduction. |
| Offspring initial HP | 3 | Initial HP of newly created offspring. |
| Physical ability (mean, std) | 6, 0 | Mean and standard deviation of agent ability. |
| Physical scaling (slope, intercept) | 5, 0.1 | Slope and intercept for ability-based interactions. |
| **Resource Parameters** | | |
| Plant: Initial quantity | 4 | Starting number of edible units per plant. |
| Plant: Capacity | 3 | Maximum capacity for plant nodes. |
| Plant: Respawn delay | 10 steps | Turns required before depleted plants respawn. |
| Plant: Nutrition | 3 | HP restored per unit consumed. |
| Prey: Initial quantity | 4 | Initial number of prey in the environment. |
| Prey: HP (mean, std) | 5, 1 | Mean and standard deviation of prey health points. |
| Prey: Physical ability | 4 | Physical ability value of prey. |
| Prey: Respawn rate | 0.1 | Probability of new prey spawning per step. |
| Prey: Max quantity | 6 | Maximum number of prey allowed in environment. |
| Prey: Difficulty | 2 | Abstract scaling factor for prey behavior/resistance. |
| Resource abundance | 2 | Global multiplier for resource density. |
| **LLM Parameters** | | |
| Provider | OpenAI | LLM provider name. |
| Model | GPT-4.1-mini-2025-04-14 | Identifier for the chat model used. |
| Max retries | 10 | Number of retries for failed LLM actions. |
| Reflection round | Enabled | Whether two-stage prompting is used. |

Table 10: This table shows the configuration parameters, their descriptions, and the values used for baseline experiments. Other experiments change only their appropriate parameters: for resource scarcity, we change the resource abundance to 1x; for high communication cost, we change the social interaction steps to 1; for moral type observability, we change the visibility of other agents' moral types to be invisible.